Biomedical Instruments Lab

STATE INSTITUTE OF TECHNICAL TEACHERS TRAINING AND RESEARCH

GENERAL INSTRUCTIONS

Rough record and Fair record are needed to record the experiments conducted in the laboratory. Rough records are needed to be certified immediately on completion of the experiment. Fair records are due at the beginning of the next lab period. Fair records must be submitted as neat, legible, and complete.

INSTRUCTIONS TO STUDENTS FOR WRITING THE FAIR RECORD

In the fair record, the index page should be filled properly by writing the corresponding experiment number, experiment name , date on which it was done and the page number.

On the **right side** page of the record following has to be written:

1. **Title**: The title of the experiment should be written in the page in capital letters.
2. In the left top margin, experiment number and date should be written.
3. **Aim**: The purpose of the experiment should be written clearly.
4. **Apparatus/Tools/Equipments/Components used:** A list of the Apparatus/Tools /Equipments /Components used for doing the experiment should be entered.
5. **Principle**: Simple working of the circuit/experimental set up/algorithm should be written.
6. **Procedure:** steps for doing the experiment and recording the readings should be briefly described(flow chart/programs in the case of computer/processor related experiments)

7. **Results**: The results of the experiment must be summarized in writing and should be fulfilling the aim.

8. **Inference** : Inference from the results is to be mentioned.

On the **Left side** page of the record following has to be recorded:

1. **Circuit/Program**: Neatly drawn circuit diagrams/experimental set up.

2. **Design**: The design of the circuit/experimental set up for selecting the components

   should be clearly shown if necessary.

3. **Observations:** i) Data should be clearly recorded using Tabular Columns.

ii) Unit of the observed data should be clearly mentioned

iii) Relevant calculations should be shown. If repetitive calculations are needed, only show a sample calculation and summarize the others in a table.

4. **Graphs :** Graphs can used to present data in a form that show the results obtained, as one or more of the parameters are varied. A graph has the advantage of presenting large

amounts of data in a concise visual form. Graph should be in a square format.

GENERAL RULES FOR PERSONAL SAFETY

1. Always wear tight shirt/lab coat , pants and shoes inside workshops.
2. REMOVE ALL METAL JEWELLERY since rings, wrist watches or bands, necklaces, etc. make excellent electrodes in the event of accidental contact with electric power sources.
3. DO NOT MAKE CIRCUIT CHANGES without turning off the power.
4. Make sure that equipment working on electrical power are grounded properly.
5. Avoid standing on metal surfaces or wet concrete. Keep your shoes dry.
6. Never handle electrical equipment with wet skin.

7. Hot soldering irons should be rested in its holder. Never leave a hot iron unattended.

8. Avoid use of loose clothing and hair near machines and avoid running around inside lab .

TO PROTECT EQUIPMENT AND MINIMIZE MAINTENANCE:

DO: 1. SET MULTIRANGE METERS to highest range before connecting to an unknown source.

2. INFORM YOUR INSTRUCTOR about faulty equipment so that it can be sent for repair.

DO NOT: 1. Do not MOVE EQUIPMENT around the room except under the supervision of an instructor.

Experiment No. 1.1

PACEMAKER CIRCUIT

Aim: To design and setup a pacemaker circuit with OPAMP IC 741C and observe the waveforms.

Objectives: After completion of this experiment, student shall be able to design and setup a circuit to simulate pacemaker using astable and monostable multivibrator.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Dual power supply +/- 15V	1
2	Function generator, 1MHz	1
3	CRO/DSO	1
4	Bread board	1
5	IC 741C	2
6	Diode 1N4007	2
7	Probes and connecting wires	As required.

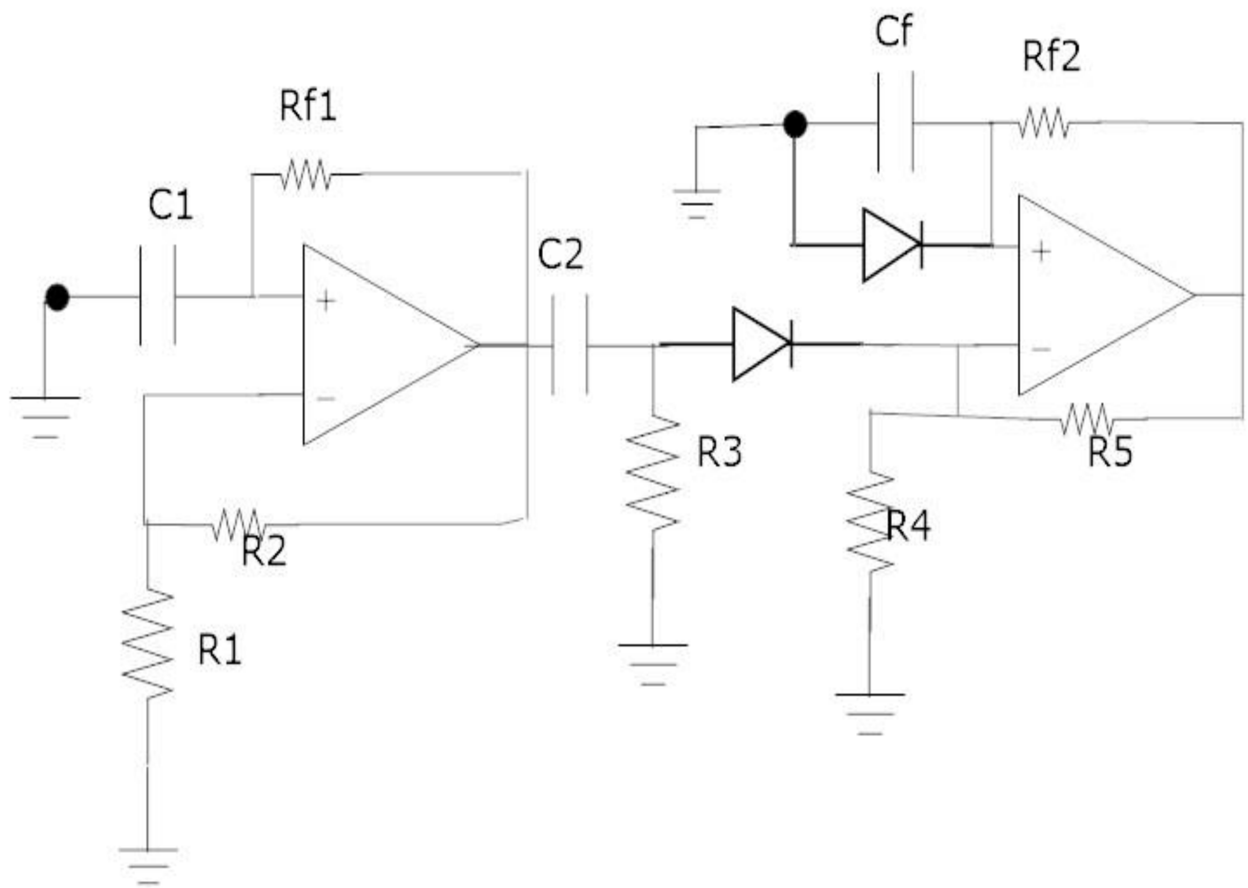
Principle:

Pacemaker circuit is used to generate pacing pulses to pace the heart artificially. The astable multivibrator produces square pulses which sets the pacing rate of 72bpm. It is then differentiated and given as trigger input to the monostable multivibrator which produces pacing pulses of suitable width.

Procedure:-

1. Check the components.
2. Setup the circuit on the breadboard and check the connections.
3. Switch on the power supply, Give Vcc as 15V.
4. Draw the output pulse waveform.

Circuit:-



Design:-
Astable:-

T= 0.83sec = 60/72
T=2Rf1C1ln[(1+ß)/(1-ß)]
ß = 0.5, C1= 0.1µF, R1=R2=1K
ie, Rf1= 3.8MΩ

Differentiator:-
R3 C2<= 0.0016T

C2=0.1µF, R3=13.2KΩ

Monostable:-

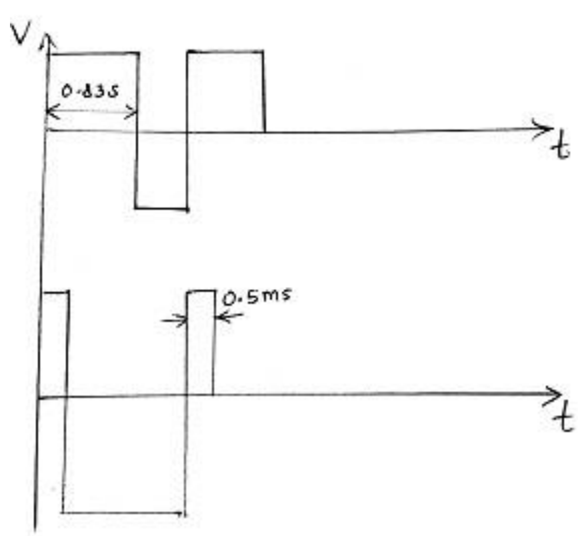
T= 0.5ms

T= Rf2 Cf ln(1/1-ß)

Cf= 0.1µF, R4=R5=1K,

Rf2= 7.2K

Graph:-



Result:

The pacemaker circuit is designed and setup successfully. Pacing pulses of width 0.5 ms are obtained for every 0.83sec.

Plot the astable and monostable waveforms.

Inference:

The pacemaker circuit can be used to give artificial pacing to a patient when in need.

Experiment No. 1.2

AUTOMATIC GAIN CONTROL

Aim: To design and setup an automatic gain control circuit with OPAMP IC 741C and observe the waveforms.

Objectives: After completion of this experiment, student will be able to design and setup an automatic gain control circuit using OP AMP. He/she will get ability to design an automatic gain control circuit

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Dual power supply +/- 15V	1
2	Function generator, 1MHz	1
3	CRO/DSO	1
4	Bread board	1
5	IC 741C	1
6	BFW 10	1
7	Probes and connecting wires	As required.

Principle:

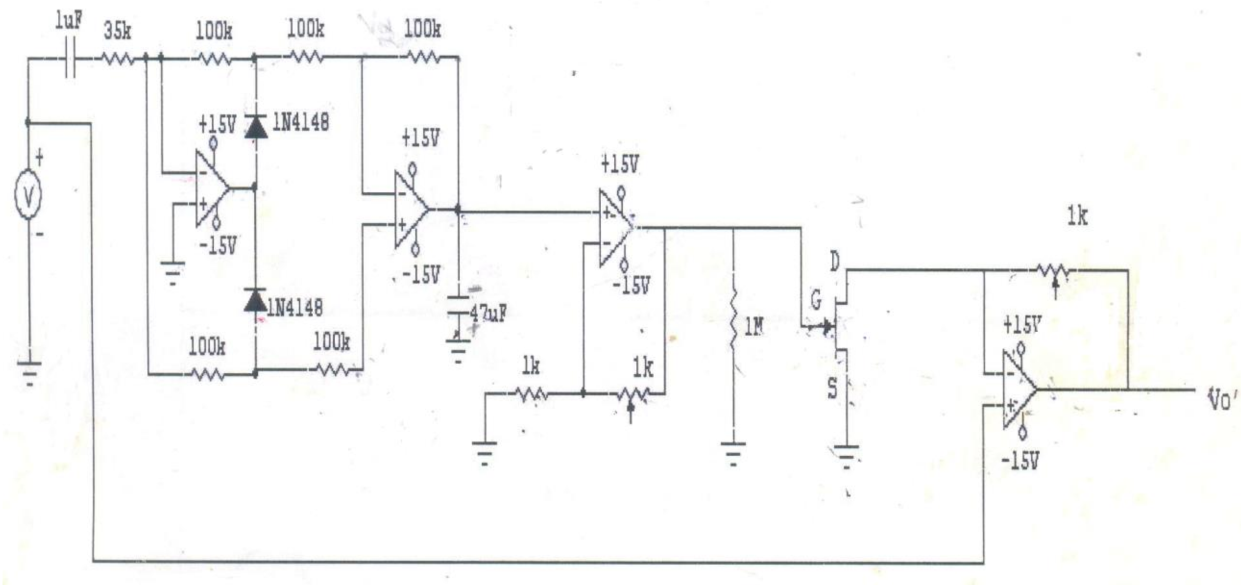
AGC is the system by means of which the overall gain of an ultrasound receiver is automatically varied with the changing strength of the received signal to make the output substantially constant. The FET is used as a voltage variable resistor. The signal is rectified using precision rectifier and its peak voltage is detected and is fed to the input of FET.

Procedure:

1. Check the components.
2. Setup the circuit on the breadboard and check the connections.
3. Switch on the power supply, Give Vcc as 15V.
4. Vary input voltage in the frequency range of 0.2 KHz to 2 KHz
5. Continuously verify the gate voltage of the FET using DMM while applying the input voltage.

6. Verify the output voltage for different levels of input voltages.

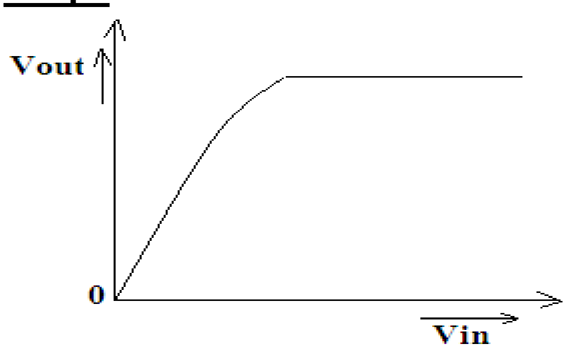
Circuit diagram



Observations

V_{in}	V_0

Graph



Result: -

The automatic gain control circuit is designed and setup successfully. Verified the output voltages for different levels of input voltages.

Plot input voltage vs. Output voltage graph

Inference:

The automatic gain control circuit can be used to increase the overall gain of the radio receiver by keeping the output substantially constant.

Experiment No. 1.3

X RAY TIMER CIRCUIT

Aim: To design and setup an X ray timer circuit with IC 555 for an exposure time of 0.2S.

Objectives: After completion of this experiment, student will be able to design and setup an Xray timer circuit with IC 555 for an exposure time of 0.2S. He/she will get ability to design an Xray timer circuit.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Dual power supply +/- 15V	1
2	Function generator, 1MHz	1
3	CRO/DSO	1
4	Bread board	1
5	IC 555	1
6	Zener diode	2
7	Transistors – 2N2222	1
8	Centre tapped transformer	9-0-9
9	Diode 1N4001	2
10	Probes and connecting wires	As required.

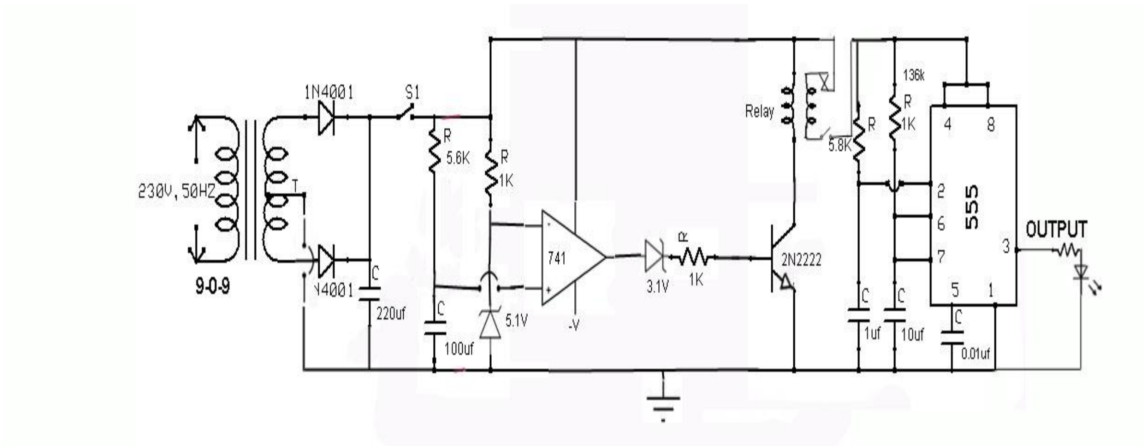
Principle:

The X-ray timer control circuit is used to control the exposure time of X-rays. The ac mains supply is rectified using full wave bridge rectifier. When the exposure switch is closed the capacitor charges through a resistor for time t, called warm up time. When the capacitor voltage crosses zener voltage, the comparator switches to positive saturation and the transistor conducts through the relay coil and normally open RLY contact gets closed. Now the remaining circuit gets power. The monostable circuit using IC 555 also gets trigger when the relay switch closes. The output of the monostable multivibrator becomes high for a time T_{ON}.

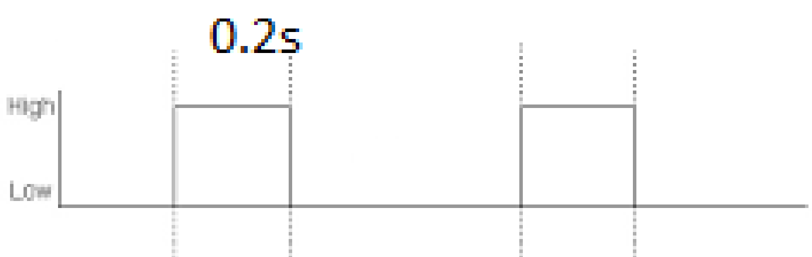
Procedure:

1. Check the components.
2. Setup the circuit on the breadboard and check the connections.
3. Switch on the power supply, Give Vcc as 10V.
4. Then setup fullwave rectifier circuit with filter. Check the output using CRO. Use a load resistor of 5K in the rectifier circuit to check the output since the input impedance of CRO is very high. The output peak should be $\sqrt{2}$ times of the rms value of the transformer output (for e.g. if you are using 9-0-9 transformer the output peak is $9*\sqrt{2}=12.7V$). Remove the load resistor and wire the capacitor charging circuit. Ensure that the capacitor is charging. Then wire the zener part of the circuit. Check the zener voltage. Next stage is comparator circuit and verify that comparator switches to positive saturation after a delay (warm up time) when the exposure switch is closed. Wire the transistor part including relay coil. Leave the relay switch not connected. If the relay holding sound is heard while closing exposure switch then connect the relay switch and check the output. After that wire the monostable section and check the output using CRO. On each time the exposure switch closes you get a low to high switching of 0.2S in the IC 555 output.

CIRCUIT DIAGRAM



Output Waveform:-



Result: -

The X ray timer circuit is designed and setup successfully.

Close the exposure switch and verify the output time and warm up time.

Draw the output of monostable multivibrator and verify ontime and offtime.

Inference:-

Using the X ray timer circuit we can control the exposure time of X ray tube.

Experiment No. 1.4

CHART DRIVE CIRCUIT

Aim: To design and setup a constant speed chart drive circuit using IC 741C .

Objectives: After completion of this experiment, student will be able to design and setup a constant speed chart drive circuit with IC 741C. He/she will get ability to design a constant speed chart drive circuit for different reference voltage.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Dual power supply +/- 15V	1
2	Variable power supply	2
3	CRO/DSO	1
4	Bread board	1
5	IC 741	2
6	DC motor	1
7	Transistors – 2N2222 , 2N3055	1 each
8	Probes and connecting wires	As required.

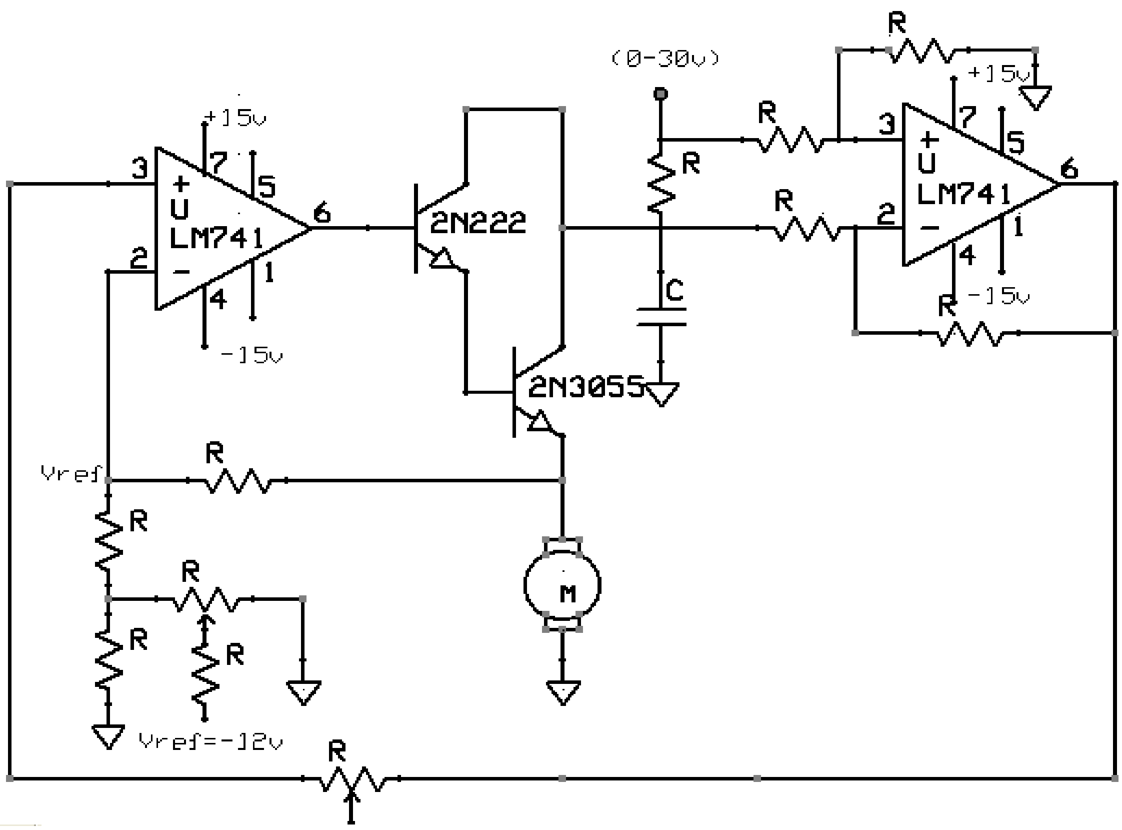
Principle: -

The circuit consists of a power amplifier for driving the dc motor, and an error amplifier for maintaining constant speed against supply fluctuations. The speed of the motor is kept constant by keeping constant voltage across the DC motor. One input to the error amplifier is a reference voltage and the other input is output of a differential amplifier whose output changes with the supply voltage variations. Error amplifier compares these two and provides a constant voltage across the DC motor.

Procedure: -

Check all the components**.** Setup the circuit on a breadboard. Ensure that the polarity of the motor is correct. Give ±15V as V_{CC} and –12V for the reference voltage circuit. Set the reference voltage, at point A, to a fixed voltage (say -5V) by adjusting the potentiometer in reference voltage circuit. Vary the input voltage from 0 to 30V and measure the voltage across the motor. Repeat the same procedure for other two reference voltages also.

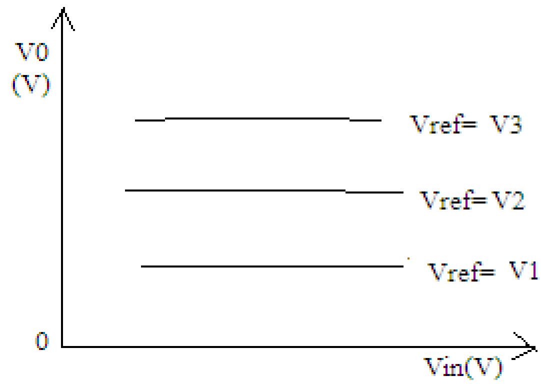
<u>Circuit</u>



<u>Observation</u>

<u>Graph</u>

$V_{ref}= V_1$		$V_{ref}= V_2$		$V_{ref}= V_3$	
V_{in}	V_0	V_{in}	V_0	V_{in}	V_0



Result: -

The chart drive circuit is designed and setup successfully. Verified the output voltages for different levels of input voltages.

Plot input voltage vs. output voltage graph.

Inference:-

Using the Chart drive circuit we can set up constant speed for different reference voltage.

Experiment No. 1.5

ESU WAVEFORM GENERATOR

Aim: To design and setup a circuit to simulate ESU waveform generator using IC 4011 and IC 555.

Objectives: After completion of this experiment, student will be able to design and setup ESU waveform generator. He/she will get ability to design a ESU waveform generator circuit.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Fixed power supply +5V	1
2	Isolation transformer	3
3	CRO/DSO	1
4	Bread board	1
5	IC 555	1
6	IC 4011	1
7	Diode IN4001	1
8	Probes and connecting wires	As required.

Principle: -

The circuit consists of a high frequency astable multivibrator and a low frequency astable multivirator. IC 4011 is a quad 2-input NAND gate. NAND gate oscillator circuit is used to generate the high frequency, 500kz and oscillator circuit using IC 555 is for the low frequency 150kHz. The square wave generated by the NAND gate oscillator is integrated and smoothened into sine wave by using pulse transformer to obtain the ‘cut waveform’.

The ‘coagulation waveform’ is generated by smoothening the wave obtained by ANDing the output of the high frequency oscillator and low frequency oscillator. The ‘blend waveform’ is obtained by combining both ‘cutting’ and ‘coagulation’ waveforms.

Procedure: -

Check all the components**.** Setup the NAND gate oscillator circuit. Give ±15V as V_{CC}. Set the output frequency to 500KHz. Then wire the integrating part of cutting waveform and check the output. After that connect the pulse transformer and if the output wave shape is distorted slightly adjust the value of capacitor to get the correct waveform.

Setup the astable multivibrator circuit. Give +15V supply as V_{CC}. Set the output frequency to 150KHz. Then connect the two NAND gates to AND the 500KHz and 150KHz wave and verify the output. Then wire the integrating part of the coagulation waveform and check the output. After that connect the pulse transformer and if the output wave shape is distorted slightly adjust the value of capacitor to get the correct waveform. By using the resistive network combine the cut and coagulation waveform and check the output. Then wire the integrating part of the coagulation waveform and check the output. After that connect the pulse transformer and if the output wave shape is distorted slightly adjust the value of capacitor to get the correct waveform.

Design:-

<u>Astable multivibrator</u>

f= 100 KHz

T_{ON}= T_{OF}=.69$R_{A/B}$C= 5us

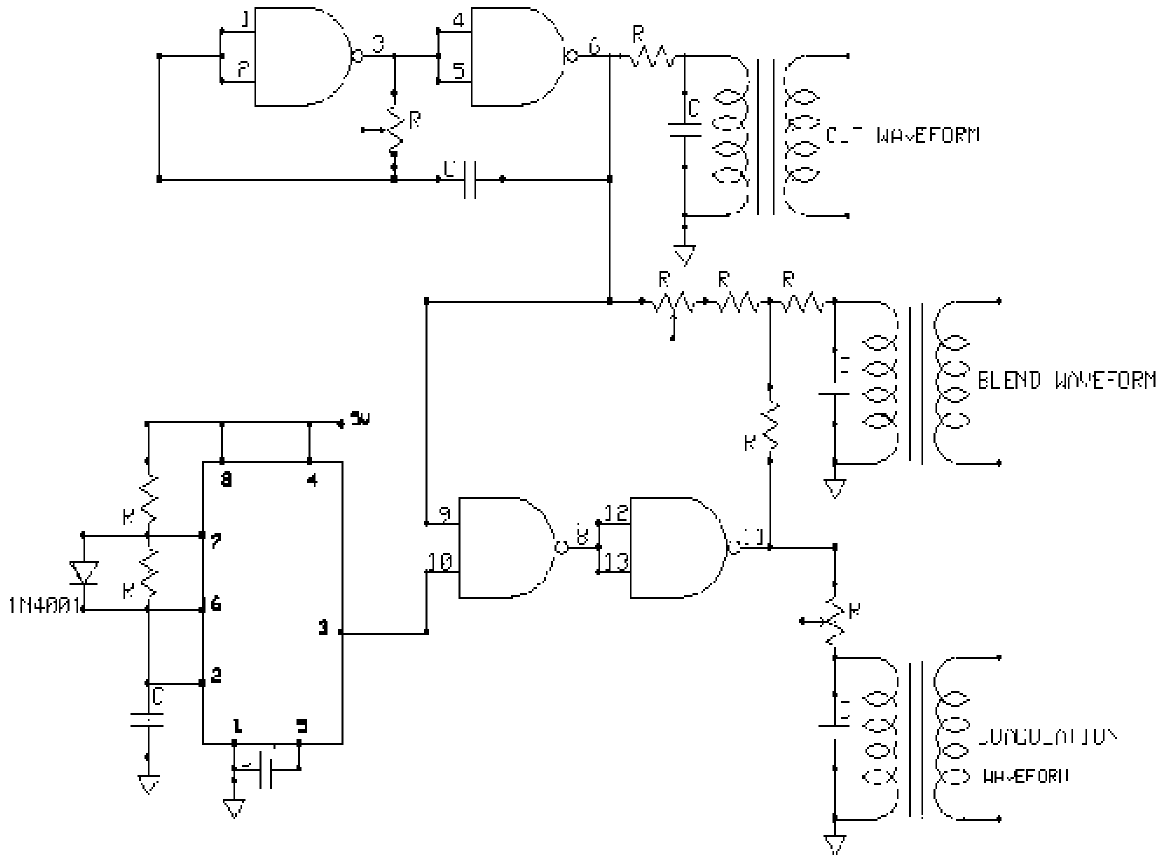
C=1uF, R=5u/.69*1uF

R_A= R_B= 7.2KΩ

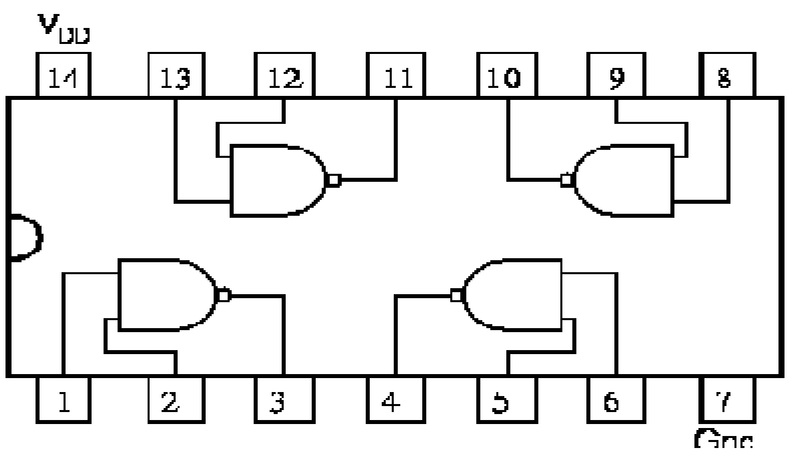
Gated AMV

F= 1/1.8RC= 500 KHz, C= 150uF , R= 7.4KΩ

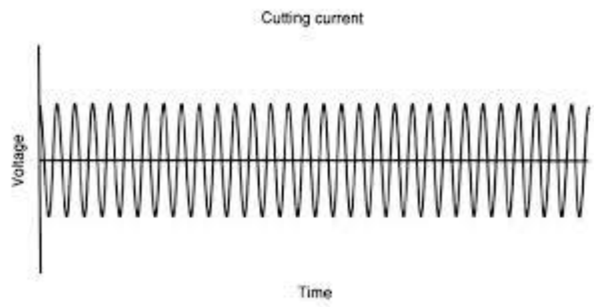
CIRCUIT

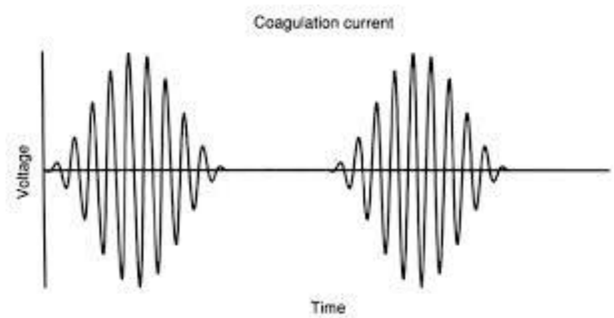


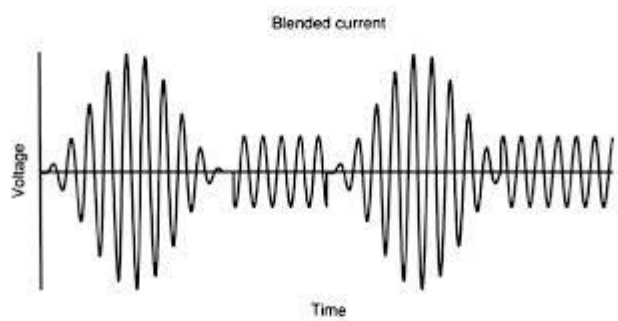
<u>4011</u>



Output Waveform:-







Result: -

The ESU waveform generator circuit is designed and setup successfully. Verified the output waveforms for different modes like cut, coagulation and blend mode.

Plot the corresponding output waveforms.

Inference:-

Using the ESU waveform generator circuit we can generate output waveforms for different modes like cut, coagulation and blend mode.

Experiment No. 1.6

ECG SIMULATOR

Aim: To design and setup a circuit to simulate the generation of ECG waveform.

Objectives: After completion of this experiment, student will be able to design and setup a circuit to generate ECG waveform. He/she will get ability to design a ECG simulator circuit.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Fixed power supply +5V	1
2	Dual Power supply +/- 15V	1
3	CRO/DSO	1
4	Bread board	1
5	IC CD4017	1
6	IC 741C	1
7	Diode 1N4148	7
8	Probes and connecting wires	As required.

Principle: -

The ECG simulator is a practical circuit used for the generation of ECG waveforms. It consists of a Johnson counter IC 4017. In this counter for successive pulses, the consecutive bit pins become high where all the other outputs remain low. The same clock is given to all the five flip-flops in the IC.

The P wave is generated during the first two clock periods. During these time interval the output Q_0 and Q_1 goes high for one clock cycle and the capacitor in the integrator starts charging. During the third clock cycle Q_2 goes high and Q_0 and Q_1 to low and the capacitor starts discharging. Required amplitude of the P wave can be obtained by adjusting the potentiometer at the output of the integrator.

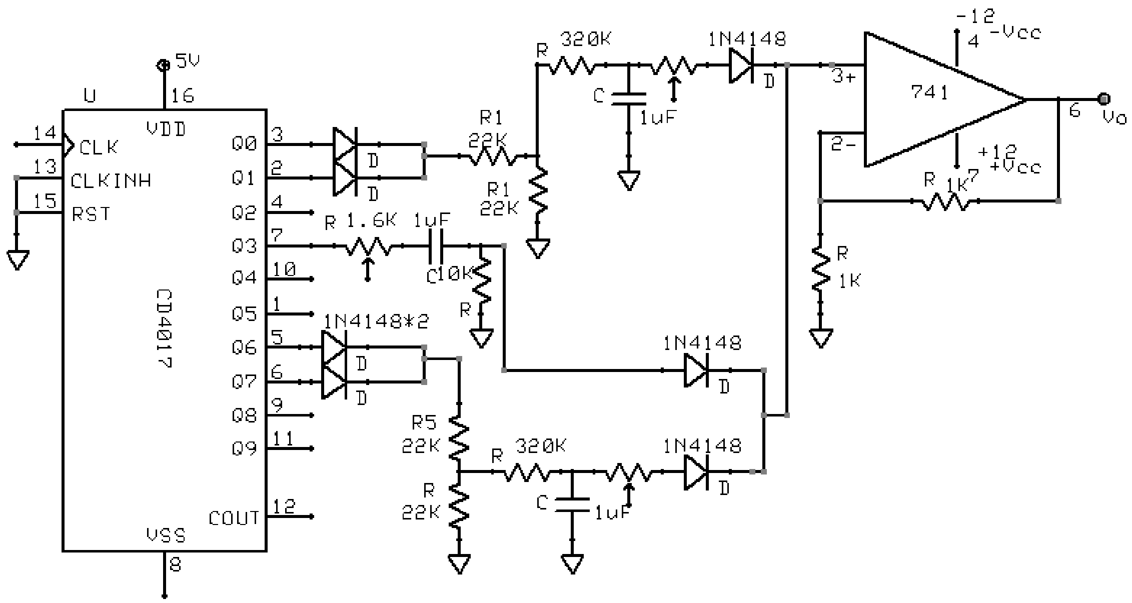
QRS wave is obtained by differentiating Q_3 output of 4017. The required amplitude can be obtained by adjusting the potentiometer used in the differentiator circuit.

The Q_6 and Q_7 outputs of 4017 are integrated to obtain T wave. The Q_2, Q_4, Q_5, Q_8 and Q_9 outputs are not used, and they provide delay between the P wave, QRS complex, and the T wave. All the three waves are applied to a summing amplifier with unity gain and the output is taken through a capacitor to obtain the correct wave shape.

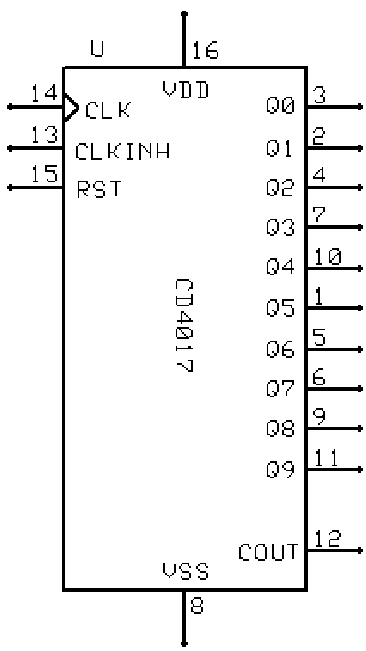
Procedure: -

Check all the components. Connect 4017 IC on the breadboard. Apply +12V as V_{CC}. Connect the Reset (pin No.13) and inhibit (pin No. 15) pins to ground. Apply a 10V, 12Hz square clock pulse from function generator in the positive pulse mode. Verify the counter output that for each successive pulse the consecutive bit pin become high and all other outputs remain low. Then wire P wave part of the circuit and verify the output and adjust the amplitude to 1V. After that wire the QRS part of the circuit. Adjust the differentiator amplitude to 1.9V. Then wire the circuit of T wave and adjust its amplitude to 1.1V. The last section is summing amplifier section. Connect all the three waves through diodes to the 3rd pin of 741 IC. Complete the circuit of summing amplifier. Take the output through a capacitor and verify the ECG output in a DSO with a time/div scale at 0.5S and volts/div at 0.5V.

Circuit diagram: -



Pin diagram of IC 4017: -



Design:-

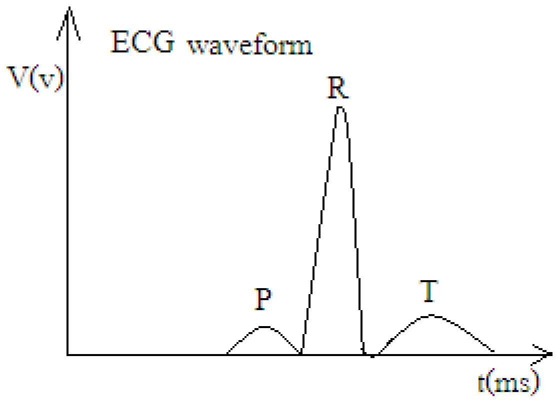
Time period of ECG signal = 0.83S
Frequency of 10 clock pulses = 1/0.83S = 1.2Hz
Input frequency = 1.2/10 = 12Hz

P-wave & T wave

P wave amplitude = 0.3V
T wave amplitude = 0.4V
RC = 16T
T = 0.2S; Let C = 1μF
R = 320K

QRS complex
QRS amplitude = 1.2V
RC = .016T
T = 0.1S; Let C = 1μF
R = 16K

Graph: -



Result: -

The ECG simulator circuit is designed and setup successfully. Verified the output waveforms for P wave, QRS complex and T wave.

Plot the corresponding output waveforms.

Inference:-

Using the ECG simulator circuit we can simulate waveforms for P wave, QRS complex and T wave.

Experiment No. 1.7

BIOTELEMETRY USING IC 4046

Aim: To design and setup frequency modulator and demodulator using PLL IC 4046 for biotelemetry.

Objectives: After completion of this experiment, student will be able to design and setup a circuit to modulate and demodulate biosignals for telemetry. He/she will get ability to design a biotelemetry circuit.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Fixed power supply +10V	1
2	CRO/DSO	1
3	Bread board	1
4	IC 4046	2
5	Probes and connecting wires	As required.

Principle: -

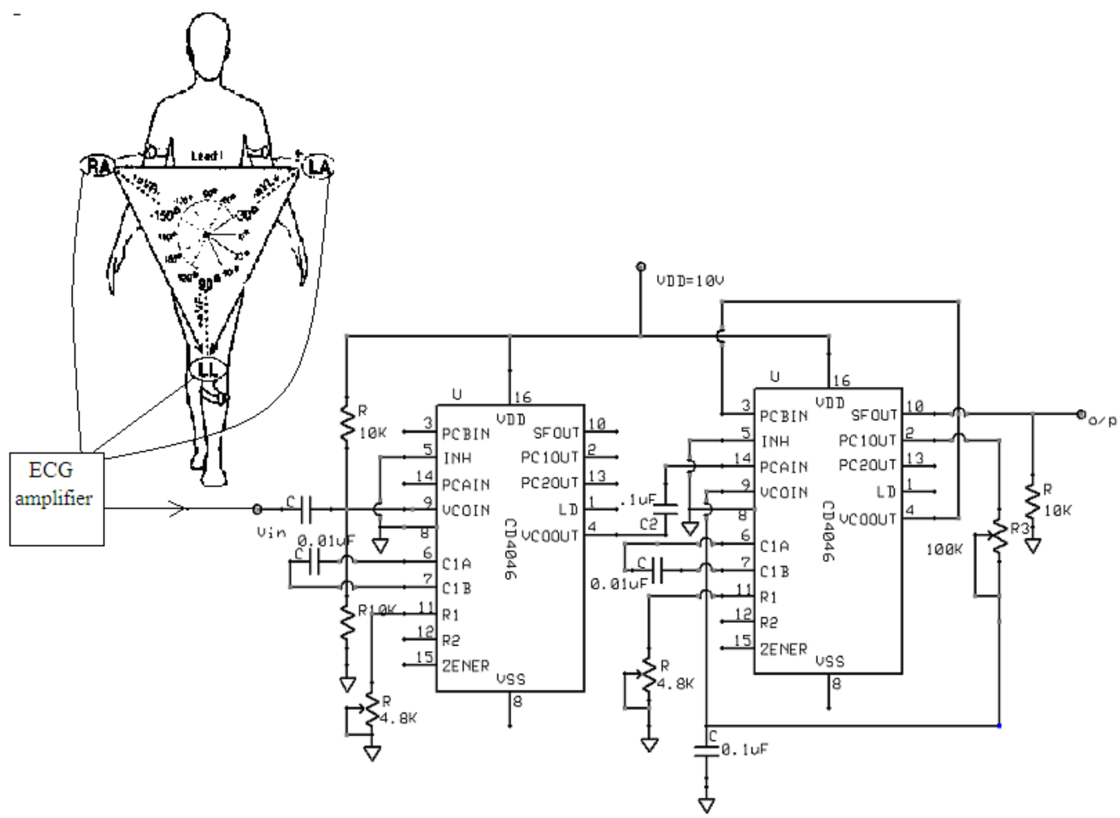
The CMOS IC 4046 is a PLL IC consisting of a low power, linear voltage controlled oscillator, two different phase comparators having a common input signal, a zener diode and a source follower. One out of the two comparators available in 4046 is an XOR phase detector while the other is an edge triggered phase detector. The XOR phase detector is most commonly used but the edge-triggered comparator is also used when high output voltage linearity is desired. An external resistor R3 and a capacitor constitute a low pass filter. A zener diode (5.2V) can be used for voltage regulation if necessary. A resistor R1 and capacitor C1determines the centre frequency of VCO, while R2 is used for offset compensation.

The VCO is concerned with modulation and low pass filter acts as a demodulator. The ECG signal to be modulated is given to the VCO, which causes a shift in the centre frequency corresponding to the amplitude of the input signal. The modulated signal is given as input to the phase comparator of a second 4046. The low pass filter blocks the high frequency carrier, thereby regenerating the original wave.

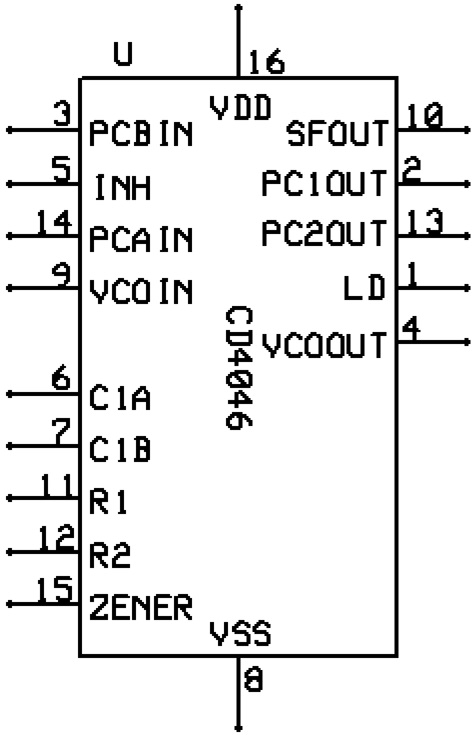
Procedure: -

Check all the components. First setup the modulator part of the circuit. Give +15V supply as V_{CC}. By connecting input point to ground set the centre frequency at 5KHz using the potentiometer R1. Disconnect the input from ground and connect ECG amplifier to the input point. Connect the Leads of the ECG amplifier in Lead III position to your body. Observe the ECG signal in a DSO with a time/div scale at 0.5S and volts/div at 0.5V. Then verify the modulated output. After that wire the demodulated part of the circuit. Verify the output is same as the ECG input in a DSO with a time/div scale at 0.5S and volts/div at 0.5V.

Circuit diagram:-



Pin diagram of IC 4046: -



Design: -

V_{DD} = 15V; f_{min} = 1Hz; f_{max} = 10KHz

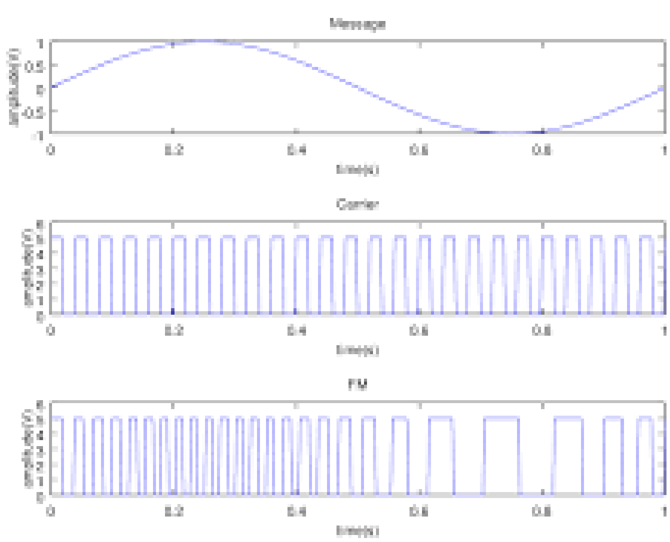
$f_0 = (f_{max} + f_{min})/2$ = 5KHz

$f_0 = 1/R_1(C_1+0.32Pf)$

Let $C_1 = 1\mu F$

R_1 = 47K

Graph:-



Result: -

The biotelemetry circuit is designed and setup successfully. Verified the modulated and demodulated output waveforms for biotelemetry.

Plot the corresponding output waveforms.

Inference:-

Using the biotelemetry circuit we can transmit biosignals successfully.

Experiment No. 1.8

QRS DETECTOR CIRCUIT

Aim:-

To design and setup QRS detector circuit using op amp IC 741C and IC 555.

Objectives:-

After completion of this experiment, student will be able to design and setup a QRS detector circuit. He/she will get ability to design a circuit to glow LED for each heart beat.

Equipments/Components:

Sl no	Name and Specification	Quantity required
1	Fixed power supply +5V	1
2	Dual Power supply +/-15V	1
3	CRO/DSO	1
4	Bread board	1
5	IC 741C	4
6	IC 555	1
7	Diode 1N4007	4
8	LED	1
9	Probes and connecting wires	As required.

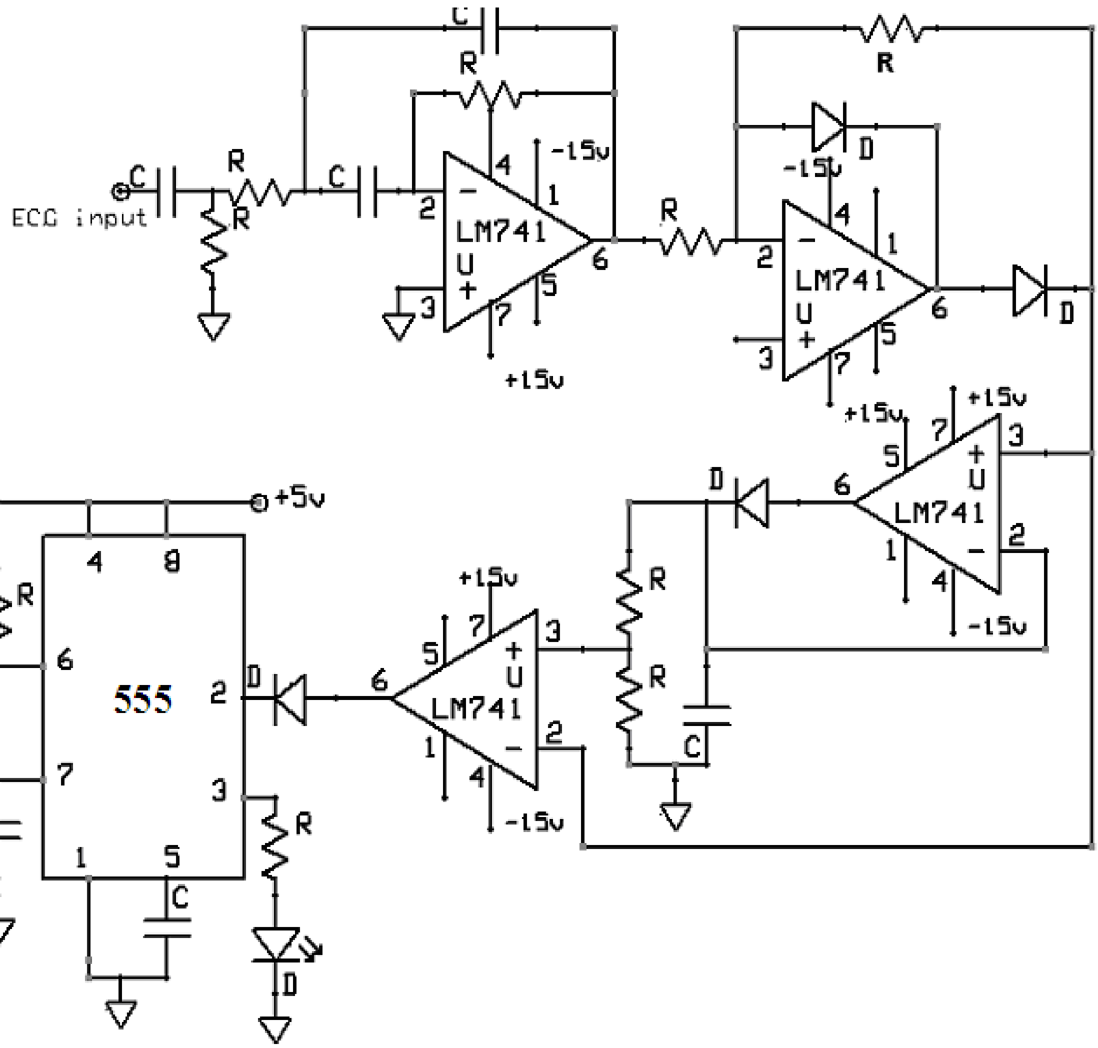
Principe: -

In this circuit, QRS complex in the ECG, acquired from patient is detected by setting an amplitude threshold. ECG from the patient is acquired using an ECG monitor and the output signal from the monitor is given to the band pass filter with a centre frequency of 17Hz and a bandwidth of 6Hz. Then peak of the voltage is detected. A portion of this peak voltage is the threshold voltage, which is derived using a voltage divider. Whenever the signal exceeds the set threshold, a QRS complex is detected and the resulting signal is used to trigger a monostable multivibrator which has T_{ON} of 0.2S. This is used to drive an LED to give a visual indication on each QRS complex.

Procedure: -

Check all the components. After that first setup band pass filter and give input to it from ECG simulator. Observe the output of the filter. Then set up the rectifier circuit and observe the output. The next stage is peak detector. Setup the circuit and check whether the peak voltage of the rectifier is getting across the capacitor. Then wire the voltage divider and verify the output ratio of voltages. Then set up the comparator circuit and check the output waveform. The Last stage is a monostable multivibrator with LED output to give visual indication on each QRS. Give 5V V_{CC} to the monostable multivibrator. Verify that the circuit is working properly. Then remove ECG simulator and connect ECG machine to give your ECG signal as the input. Connect ECG leads in Lead I mode and take output from the back panel 25 pin connector (between pin 10 & pin 11) of the ECG monitor.

Circuit diagram:-



Design:-

Bpf

fc =17Hz

BW =6Hz

Q = fc/BW= 17/6= 2.83

A=2

C1=c2= .1uF

V=5V

I= 20mA, R=5/20mA= 250Ω

R1= $Q/2\pi f_c A_p$= 150KΩ

R2= $Q/2\pi f_0\,(2Q^2 - A_p)$=22KΩ

Monostable

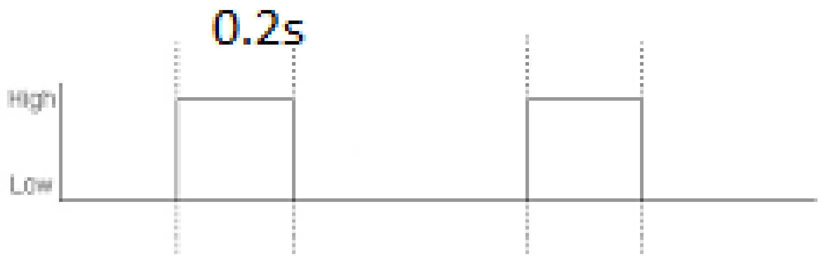
T=1.1RC

T=0.2S

R=100K

C=0.2/1.1*100K= 1.8uF

Output waveform of monostable



Result: -

The QRS detector circuit is designed and setup successfully. Verified the output waveforms at each stage.

Plot the corresponding output waveforms.

Inference:-

Using the QRS detector circuit we can calculate heart rate by counting LED display for 1 minute.

Experiment No. 2.1

ECG RECORDER

Aim:- To demonstrate the working of ECG monitor/recorder.

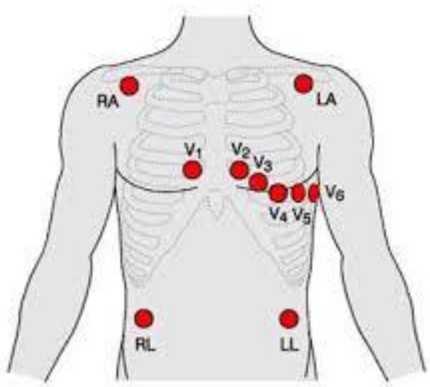
Objectives:- After completion of this experiment, student will be able to demonstrate the working of ECG recorder

Principle:-

The electrocardiogram (ECG) is a graphical record of electric potentials generated by the heart muscle during each cardiac cycle. These potentials are detected on the surface of the body using electrodes attached to the extremities and chest wall, and are then amplified by the electrocardiograph machine and displayed on special graph paper. ECG Machine consists of the ECG unit, electrodes, and cables. The 12-lead system includes three different types of leads: bipolar, augmented and unipolar. Each of the 12 standard leads presents a different perspective of the heart's electrical activity; producing ECG waveforms in which the P waves, QRS complex, and T waves vary in amplitude and polarity. Single-channel ECGs record the electric signals from only one lead configuration at a time, although they may receive electric signals from as many as 12 leads. Non interpretive multichannel electrocardiographs only record the electric signals from the electrodes (leads) and do not use any internal procedure for their interpretation. Interpretive multichannel electrocardiographs acquire and analyse the electrical signals.

Procedure:-

1. Patient Preparation:-
   - 1.1 Skin Preparation – Shave hair from electrode sites. Cleanse electrode sites with surgical spirit to remove dirt or oil. Lightly abrade skin at electrode site with a skin rasp. The top layer of epidermis has to be removed.
   - 1.2 Cable routing – Ensure all leads are properly routed and are not tangled with each other.
   - 1.3 Electrode placement – Place the disposable electrodes at the locations shown in the figure:



Electrode Name	Colour	Position	System
RA	White ('snow')	Right arm	3-elctrode 5-electrode 12-lead ECG
LA	Black ('smoke')	Left arm	3-elctrode 5-electrode 12-lead ECG
LL	Red ('fire')	Left leg	3-elctrode 5-electrode 12-lead ECG
RL	Green ('grass')	Right leg	5-electrode 12-lead ECG
C	Brown	Central chest Over sternum	5-electrode
V1	Red	Sternal edge Right 4th ICS	12-lead ECG
V2	Yellow	Sternal edge Left 4th ICS	12-lead ECG
V3	Green	Between V2 and V4	12-lead ECG
V4	Blue	Mid-clavicular line Left 5th ICS	12-lead ECG
V5	Orange	Between V4 and V5 Left 5th ICS	12-lead ECG
V6	Purple	Mid-axillary line Left 5th ICS	12-lead ECG

Main machine components:-

1. Acquisition device
2. ECG lead patient cable
3. Reusable ECG electrodes
4. Thermal paper role
5. Paper role axle
6. ECG jelly
7. Battery charger
8. Fuse

Result: -

The working of ECG recorder is demonstrated successfully.

Inference:-

Using the ECG recorder ECG of a patient can be recorded successfully.

Experiment No. 2.2

EMG RECORDER

Aim:- To demonstrate the working of EMG recorder.

Objectives:- After completion of this experiment, student will be able to demonstrate the working of EMG recorder

Principle:-

Following testes can be done on this machine

1. BAER TEST (Brain stem Auditory Evoked Potential)
2. MNCV (Motor nerve conduction velocity)
3. SNCV (Sensory nerve conduction velocity).
4. RNS (Repetitive nerve stimulation).
5. F-Wave test.
6. Blink reflex test.
7. EMG test.
8. VEP (Visual evoked potential)
9. SSEP (Somato sensory Evoked Potential)

Procedure: -

Switch on the adapter box of EMG. Make double click on the EMG. Click on MNC menu. Click on pt information icon or press F9 on keyboard to feed patient information. Hence clear field button will clear all fields except physician and technician name. After feeding patient information save it and click OK. Click on new settings icon at bottom of screen or press <F7> on keyboard, press default settings button and adjust settings like amplifiers, sweep speed sensitivity, rise time, delay, threshold etc. Now click on the OK button to save changes. Click on Start New Test button for new test. Select the nerve for testing. Apply the electrodes on appropriate place. Adjust the current level on the shock stem handle. Deliver shock pulses by pressing on switch on the handle select the capture icon. For measuring MNC two different locations are selected for delivering shock pulses. Measure the distance between two shock pulses and enter the value. Conduction velocity is displayed on the screen.

Specification& power requirements

Computer System Minimum Configuration:

- CPU minimum PIII 633 MHz or above.
- 10GB HDD.
- RAM 64MB SD RAM.
- Color Monitor 15"Supporting 1024 x 768 resolution.
- 1.44MB FDD.
- Keyboard 104 keys.
- Printer Laser/DeskJet

Power Requirement:

- 220V ±10V A.C.
- Minimum Watts: 650 W.
- The power supply should have proper grounding (voltage between ground and neutral should be less than 2 volts).

Specification & Power Requirements

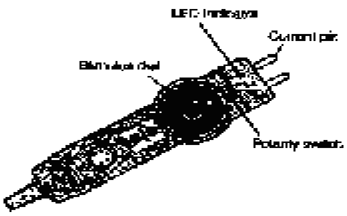
1. A wash basin must be installed in the installation room
2. To eliminate electromagnetic interference, should not be located near transformers, D.C. motors or other power appliances.
3. Do not connect on the same phase that is used for air-conditioner.
4. Grounding of the AC outlets is mandatory.
5. Wire/cables carrying large current should not pass through installation room.
6. The installation room should have separate grounding.

Standard Items :		**<u>EMG Sub-Assemblies</u>**
1.	Adaptor Box	01
2.	Foot Switch	01
3.	Shock Stim Handle	01
4.	Head Phone	01
5.	USB Interface box	01
6.	AM-AM USB Cable	01
7.	16 pin flat cable	01
8.	Amplifier 2 ch. (4 Ch. optional)	01
9.	Serial Cable	01
10.	Installation CD	01
11.	Spike Protector	01
12.	EMG Cart with Amplifier Arm	01
13.	Dust Cover for Complete Machine	01
14.	Electrodes - (Surface, Sensory, EP, EMG Needle Cable, Ground electrode, Concentric Needles)	01 set each
15.	Jumper Cable	05
16.	Paste Jar	01
17.	K-Y-Jelly	01

1.	CPU	01
2.	Color Monitor	01
3	Keyboard	01
4.	Mouse & Mouse Pad	01
5.	Power Cable	01
6.	Speakers	01 set
7.	Printer (Laser B/W)	01
8.	Printer Interface Cable & CD for installation printer	01 each
9.	VEP Monitor (monochrome 14")	01

1. Adaptor Box

2. Foot Switch

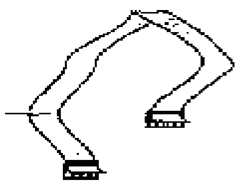


3. Shock Stim Handle

4. Head Phone

5. USB Interface box

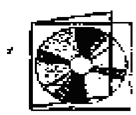
6. AM - AM USB Cable



7. 16 pin flat cable

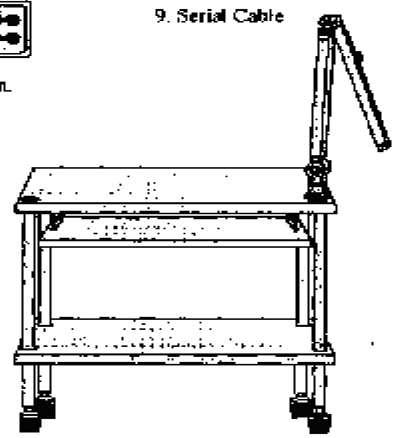
8. Amplifier 2 ch.

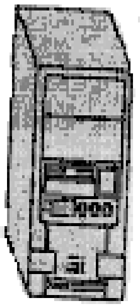
9. Serial Cable



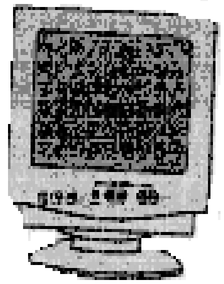
10. Installation CD

11 Spike Protector



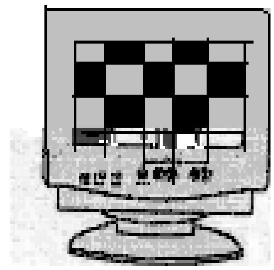


1.CPU

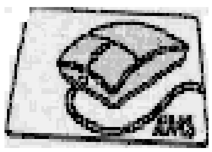


2.Color Monitor

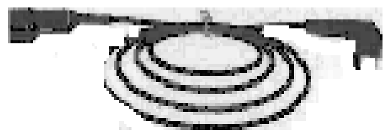
3.Keyboard



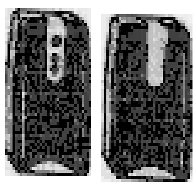
9. VHP Monitor



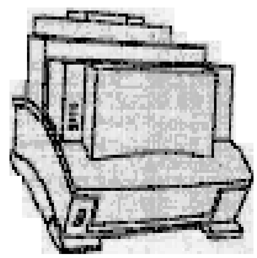
4. Mouse & Mouse Pad



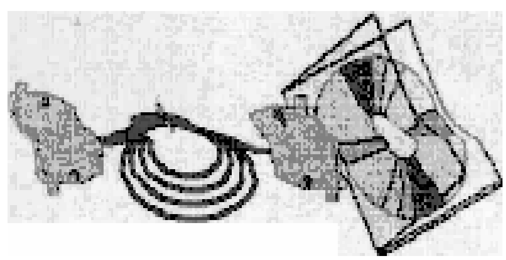
5. Power Cable



6.Speakers

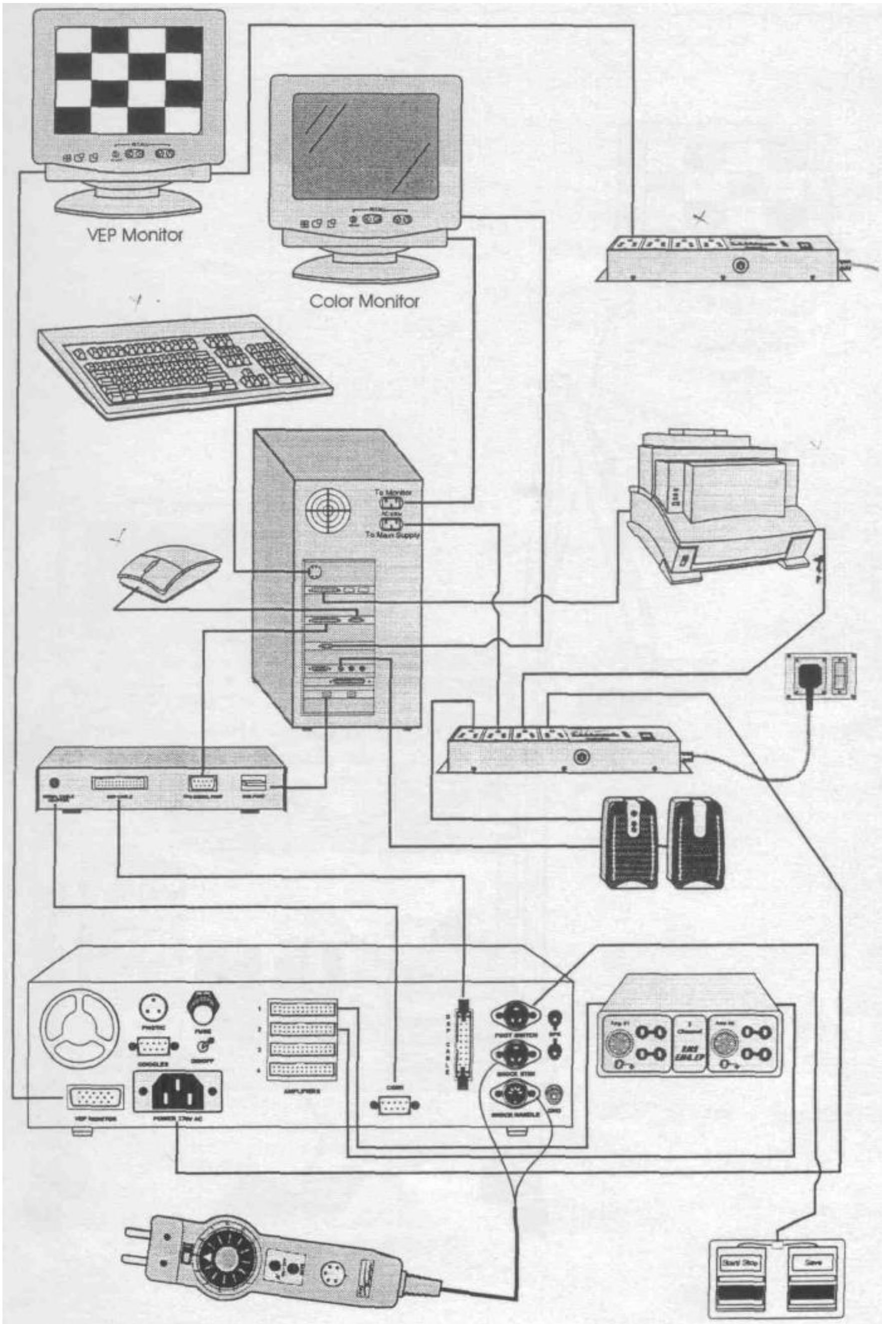


7. Printer (Laser BAV)

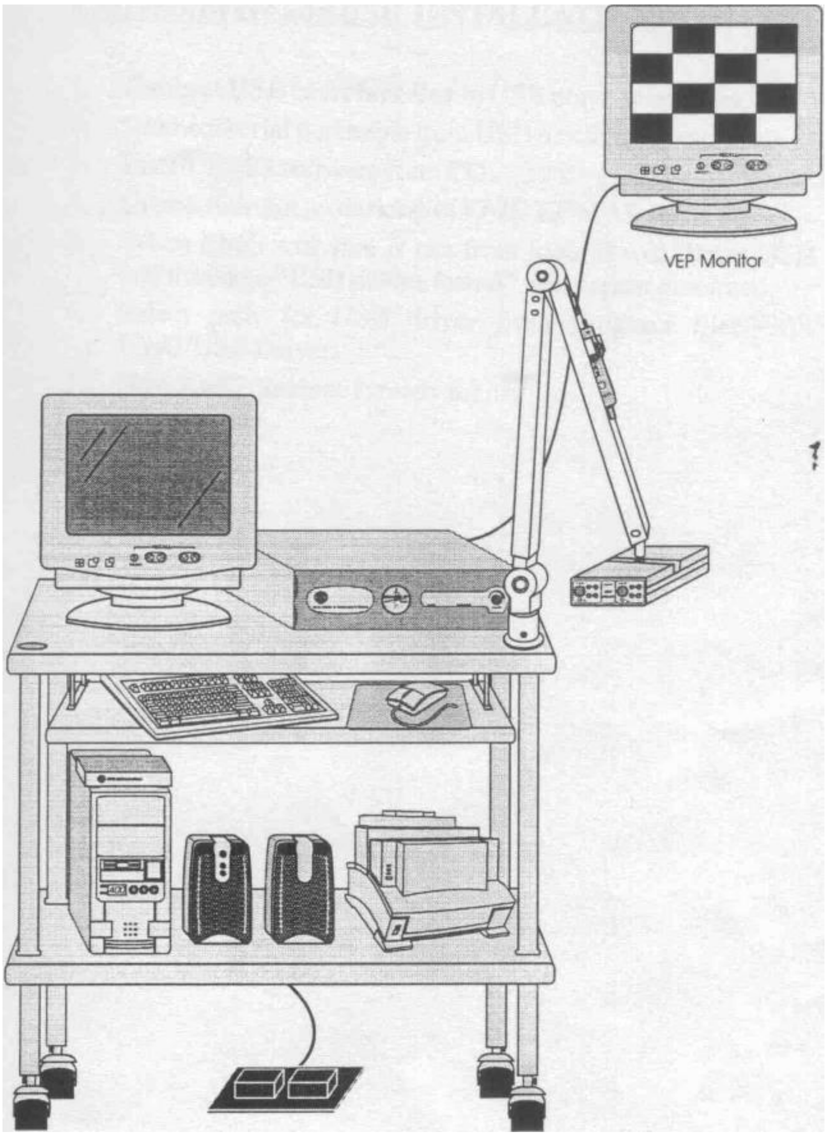


8. Printer Interface Cable & Printer Driver CD

INTERCONNECTION DIAGRAM 1



INTERCONNECTION DIAGRAM 2



Result: -

The working of EMG recorder is demonstrated successfully.

Inference:-

Using the EMG recorder the EMG of a patient can be recorded successfully.

Experiment No. 2.3

EEG RECORDER

Aim:- To demonstrate the working of EEG monitor/recorder.

Objectives:- After completion of this experiment, student will be able to demonstrate the working of EEG recorder and record the EEG of a patient.

Principle:-

Brainwaves are naturally occurring electrical signals in the brain which are caused by the neurons 'firing'. These pulses can be measured on the surface of the head using an EEG. There are five distinct frequency ranges of brainwaves that indicate different brain (or mind) states. In ascending frequency order these are known as delta, theta, alpha, beta and gamma and they are correlated to different levels of awareness, focus and excitation in the brain.

An EEG device records electrical signals from the brain, specifically postsynaptic potentials of neurons originating from the cerebral cortex, through electrodes that are attached to the subject's scalp.

The electrodes attached to the subject's scalp transmit the electrical signals produced by the brain to the EEG monitor. Since these electrical signals are very small (of the order of 10s of microvolts) the EEG acts as an amplifier, typically amplifying them by 10,000 times, as well as a device to measure them.

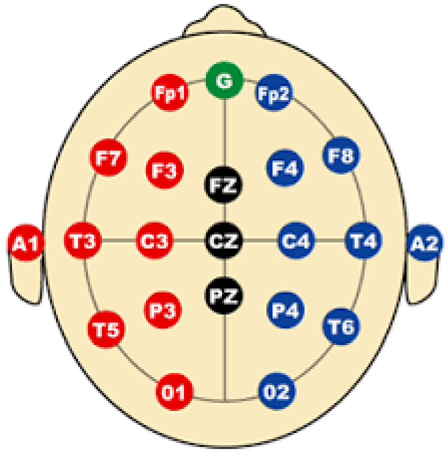
Procedure:-

Patient preparation

1. Wash the hair with shampoo or shave the hair from the scalp and wash it.
2. Remove all metallic ornaments .

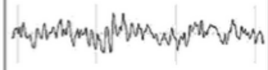
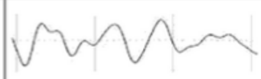
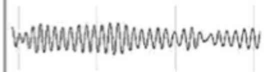
Placement of electrodes.

The internationally standardized 10-20 system is used to record the spontaneous EEG. In this system 21 electrodes are located on the surface of the scalp.



Reference points are nasion, which is the delve at the top of the nose, level with eyes; and inion, which is the bony lump at the base of the skull on the midline at the back of the head. From these points, the skull perimeters are measured in the transverse and median planes. Electrode locations are determined by dividing these perimeters into 10% and 20% intervals. Three other electrodes are placed on each side equidistant from the neighbouring points.

Equipment:-

Frequency Band Name	Frequency Bandwidth	State Associated with Bandwidth	Example of Filtered Bandwidth
Raw EEG	0–45 Hz	Awake	
Delta	0.5–3.5 Hz	Deep Sleep	
Theta	4–7.5 Hz	Drowsy	
Alpha	8–12 Hz	Relaxed	
Beta	13–35 Hz	Engaged	

Result: -

The working of EEG recorder is demonstrated successfully.

Inference:-

Using the EEG recorder EEG of a patient can be recorded successfully.

Experiment No. 2.4

SPHYGMOMANOMETER

Aim:- To demonstrate the working of sphygmomanometer.

Objectives:- After the completion of this experiment, student will be able to demonstrate the working of sphygmomanometer and measure the blood pressure of a patient.

Principle:-

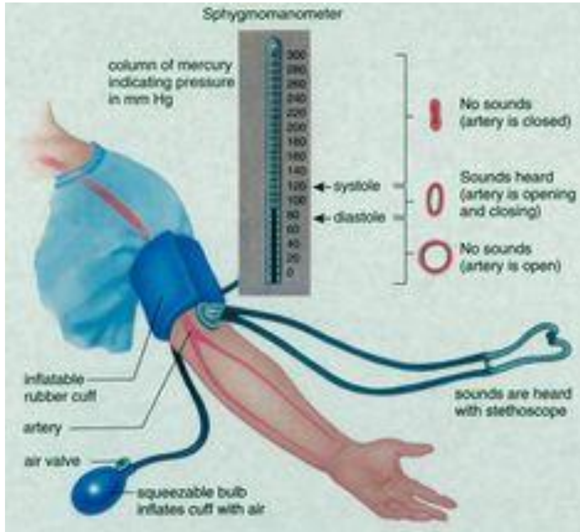
A sphygmomanometer is a medical instrument used to measure arterial blood pressure. A sphygmomanometer consists of a pump, dial, cuff and a valve. To measure blood pressure, the cuff is wrapped around the arm and then inflated using the pump. The pressure applied by the cuff closes off the brachial artery so that no blood flows through. The pressure is then slowly released and sounds are detected using a stethoscope placed on the brachial artery. Blood pressure readings are given as a fraction of two numbers, 120/80 mm Hg for example. The first number is the systolic pressure; this is the blood pressure when the ventricles of the heart are contracting. The number is recorded when a sound is heard for the first time as pressure in the cuff is slowly relaxed. The second number is the diastolic pressure, which is the blood pressure in between contractions when the heart is relaxed. This number is recorded when the thumping of blood squeezing through the narrowed brachial artery is no longer heard. It is important to note that multiple measurements of a person’s blood pressure must be made in order to gain an accurate reading.

Procedure:-

The center of the sphygmomanometer bladder should be placed over the brachial artery The lower border of the cuff should be about 2cm proximal to the antecubital fossa and the cuff should be firmly wrapped around the arm.

Figure show the position of checking the brachial artery. The subjectwill be asked to relax and sits on a chair with the lower arm supported The blood pressure cuff is placed on the subject's right arm, allowing 1 inch between the bottom of the cuff and the crease of the elbow diaphragm is placed over the brachial artery in the space between the bottom of the cuff and the crease of the elbow. At this point no sounds should be heard.The cuff pressure is inflated quickly to a pressure about 30 mm Hg higher than the systolic pressure determined by the method of palpation. Then the air is let out of the cuff at a rate such that cuff pressure falls at a rate of about 5 mm Hg/sec. At some point the person listening with the stethoscope will begin to hear sounds with each heartbeat. Then the pressure in the cuff is slowly released.

When blood starts to flow in to the artery, the turbulent flow creates a pulse synchronic pounding (first Korotkoff sound).Korotkoff sound is the arterial sounds heard through a stethoscope applied to the brachial artery distal to the cuff of a sphygmomanometer that change with varying cuff pressure and that are used to determine systolic and diastolic blood pressure.The pressure at which this sound is first detected is the systolic blood pressure. The cuff pressure is further released until no more sound canbe detected at the diastolic arterial pressure.The laminar flow that normally occurs in arteries produces little vibration of the arterial wall and therefore no sounds.However, when an artery is partially constricted, blood flow becomes turbulent, causing the artery to vibrate and produce sounds.When measuring blood pressure using the auscultation method, turbulent blood flow will occur when the cuff pressure is greater than the diastolic pressure and less than the systolic pressure.The "tapping" sounds associated with the turbulent flow are known as Korotkoff sounds.These sounds are not same with the heart sounds produced by the opening and closing of the heart valves



Result: -

The working of **s**phygmomanometer is demonstrated successfully.

Inference:-

Using the **s**phygmomanometer the blood pressure of a patient can be measured successfully.

Experiment No. 2.5

COLORIMETER

Aim:- To demonstrate the working of Colorimeter.

Objectives:- After the completion of this experiment, student will be able to demonstrate the working of colorimeter.

Principle:-

A broadband LED fed from ac IC stabilized power supply forms the light source. The intensity is controlled by an aperture disc and collimated by a lens system. This light passes through the selector filter and then on to the test tube falls on a sensitive , hermetically sealed photocell. The photocell output is amplified by a single stage IC amplifier and is readout by a log DPM. The DPM displays optical density from 0.00 to 1.99 with negative display and overrange indication.

Procedure:-

1. Set the aperture control to minimum .
2. Connect the instrument to a 230V 50Hz grounded outlet. Switch on the instrument by means of switch provided on the back.The display will read 1.This is equivalent to 0.0 on transmission scale. Wait a few minutes for the instrument to stabilize .
3. Select the filter by rotating the turret.
4. Insert a test tube containing a blank solution in the sample holder.
5. Adjust the aperture control until the display reads zero.
6. Remove the blank solution, insert the test sample and note the display readings.

Selection of Filter:-

Selecting the proper filter and obtaining reproducible readings are the principle conditions in the measurement process.

Filter	Peak(nm)	colour	Pass band
65	650	Red	6300 A to Infrared
59	590	Yellow	5500 A to Infrared
54	540	Green	4500 A to 6500 A
49	490	Blue	3500 A to 6000 A
42	420	Violet	3400 A to 5500 A

Result: -

The working of colorimeter is demonstrated successfully.

Inference:-

Using the colorimeter analysis of blood samples can be done.

Experiment No. 2.6

PH METER

Aim:- To demonstrate the working of pH meter.

Objectives:- After the completion of this experiment, student will be able to demonstrate the working of pH meter.

Principle:-

If two solutions are separated by pH glass , an electrical potential will be developed across the membrane. If the solution inside the bulb contains hydrogen ion concentration, the membrane potential will change as the hydrogen ion concentration of the other solution varies. If electrical connections are made to these solutions (a) inside the glass by pH electrode's internal element and (b) outside the glass by a reference electrode, the membrane potential can be measured by a high impedance voltmeter.

A convenient scale for the reporting of hydrogen ion concentration is derived from:

pH = $-\log_{10}$ [H+] where [H+] is the hydrogen ion activity.

The value of the unknown solutions may be presented directly in pH units or milli volts.

Procedure:-

1. Connecting the electrode
   1.1 Set up the electrode stand and fit the pH electrode into it.
   1.2 Carefully remove the procreative rubber cap from the filling hole of electrode.
2. Preparation of buffer solutions.
   Dissolve one buffer tablet or powder pack of 7 pH in 100ml distilled water. The pH of this solution is 7pH. Similarly , buffer solutions of other value can be prepared.
3. Calibration of electrode
   The electrode should be calibrated before taking measurements.
   3.1 Connect combination pH electrode to the input socket, wash it with water and switch on the instrument.
   3.2 Dip the electrode in 7pH buffer solution.
   3.3 Set the TEMPERATURE ^{0}C controls to the buffer solution temperature from the back panel.
   3.4 Set the mode selector switch to pH position and push the 'Set 7pH' control till the digital display shows the precise pH value of the buffer solution.
   3.5 Now move the Function Selector switch to STAND BY

3.6 Remove the electrode from the buffer solution and wash it with distilled water. Dip the combination electrode into another buffer solution (say 4 pH)
3.7 Set the TEMPERATURE ^{0}C controls to the temperature of selected buffer solution.
3.8 Set the Function Selector switch to pH position. “Set 4 pH” correction control at the front panel until the display shows the pH value of the selected buffer solution.

4. mV measurement.
To measure millivolts,
4.1 Connect the combination pH electrode to the input socket.
4.2 Set the Function Selector switch to mV position.
4.3 Dip the combination electrode in solution under test.
4.4 The display will display millivolt value of the solution under test.

Front Panel Controls:-

1. Mode control switch
   1.1 ATC (Autotemperature): At this position the instrument measures pH with automatic temperature compensation.
   1.2 pH: At this position , the instrument measures directly in pH units.
   1.3 STAND BY: On this position the instrument will indicate 000 +-1.
   1.4 mV: On this position the instrument will indicate millivolts from +-1999
2. Digital Display: A $3_{1/2}$ digit display that reads upto +-1999 .
3. Set 7 pH : This control is used to set 7pH on digital display when the electrode is dipped in 7pH buffer solution
4. Set 4pH: This control is used to set 4pH on digital display when the electrode is dipped in 4pH buffer solution

Back panel Controls:-

1. Input socket for temperature :- This input socket accepts the combined pH and mV electrode with BNC connector.
2. Input socket for temperature :- This BNC connector socket accepts Probe Temp. probe supplied with the instrument.
3. ON/OFF Switch: This two position switch is used to switch ON/OFF the instrument.
4. Fuse(100mA):- The 100 mA fuse is used to control the current from the power supply to the instrument.
5. Temperature Knob: This control compensates for the slope versus temperature characteristics of the electrode and operates in the pH mode only.

Result: -

The working of pHmeter is demonstrated successfully.

Inference:-

Using the pHmeter, pH of blood samples can be found out.

Experiment No. 2.7

HB METER

Aim:- To demonstrate the working of HB meter.

Objectives:- After the completion of this experiment, student will be able to demonstrate the working of HBmeter.

Principle:-

The measurement of haemoglobin concentration is carried out at wavelength of 546nm. Using LED technology green light produced is projected through the sample and measured by sensitive photodiode. The measurements are made using Cyanmethhaemoglobin method .

Procedure:-

1. Allow 5minutes warm up period after switching on the instrument.
2. Display shows “1”, Note that both the standard and sample LED’s are off.
3. Take a blank solution in a test tube and insert test tube in the holder. Make sure that mark on the test tube coincides with that on the panel.
4. Press ZERO key : The display shows 00.0 and the sample LED will glow.
5. The instrument retains the last calibration done in the memory. Put the test tube containing sample in the test tube holder and the instrument gives the result directly in Hb units.

CALIBRATION OF THE INSTRUMENT

1. In the sample measurement mode, when ‘Std’ key is pressed , the ‘Std’ LED glows.
2. Display blinks last stored value.
3. Insert test tube containing standard solution in test tube holder. The mark on the test tube should coincide with that on the panel.

4. Press ‘ENTER’ key to accept blinking standard value or press **^/ZERO** key to change standard solutions value.
5. On pressing **^/ZERO** key the first digit of the display glows. Use **^ or ˇ** keys to select the first digit and press ‘ENTER’ Key.
6. On pressing ENTER key instrument calibrates the standard internally and goes to sample measurement mode.
7. Now remove the test tube containing standard solution from the holder.
8. Put the test tube containing sample in the holder and the instrument shows the result directly in hB units.

User Controls:

Digital Display: This 3 digit seven segment red LED display is for accurate and fast reading of hB.

Test tube Holder: The test tube holder holds the test tube containing solution to be tested.

Shutter Control: In-built in the sample compartment to avoid fatigue of photodiode.

ON/OFF switch: used to switch On/OFF the instrument

ENTER: This key when pressed indicates the stored standard hB value and to enter new standard value.

ZERO: This key is used to set blank solution value to 00.0.

STD LED: This indicates that instrument is in standard mode. In this mode user can set new standard value and calibrate the instrument.

Sample LED: This indicates that the instrument is in sample measurement mode.

Result: -

The working of HBmeter is demonstrated successfully.

Inference:-

Using the HBmeter , Hb of blood samples can be done.